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Summarizing Data

Location, variability, and outliers

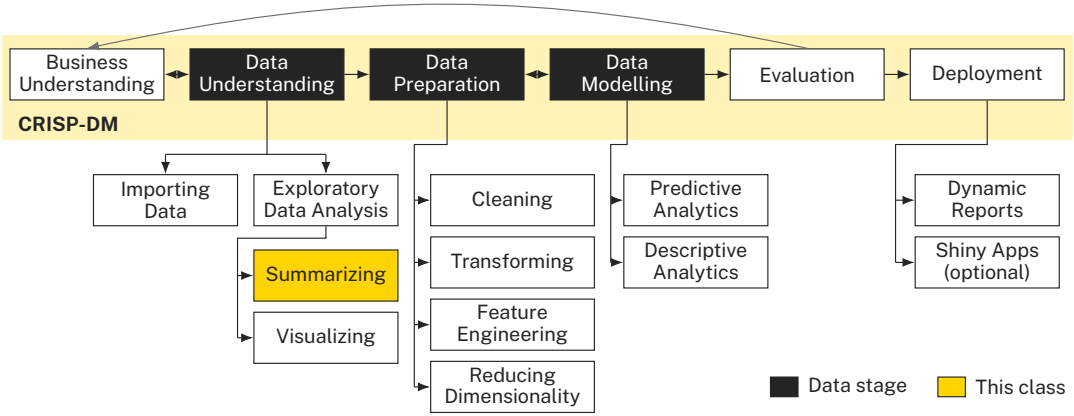
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Submit Privately, Then Share

- Lab 01 is individual work, due September 17 before 16:05 Halifax time.
- The best 10 of 12 labs count for **15%**.
- Keep the repository and cloud notebooks private, with instructor access.
- Record the commit SHA and URL in `labs/lab-01/README.md`.
- After the deadline and release, publish that commit and its history in your portfolio and present it at the next class meeting, if one follows.

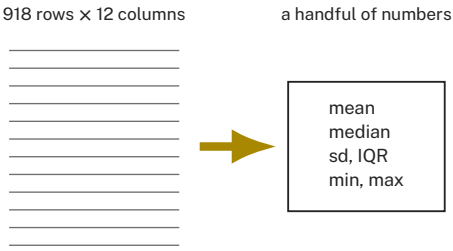
No rewriting or backdating assessment history. Corrections use new commits. Generative AI output is prohibited on labs.

Where Summarizing Sits



Why Summarize Data?

- Data are often too large for us to be able to understand what is going on just by perusing them.
- Data summaries give us an overview of key data properties.
- We want to describe the **distribution** of the observed values.



Types of Variables

- **Numeric**
  - Discrete, for example the number of visits
  - Continuous, for example temperature
- **Categorical**
  - Nominal, such as blood group; ordinal, such as low, moderate, high

Storage types do not determine analytical roles. Nominal data have no meaningful mean.

R

```r
x <- c(2, 4, 4, 5, 7, 9, 12, 41)
class(x)   # "numeric"
str(x)
# num [1:8] 2 4 4 5 7 9 12 41
```

Python

```python
import pandas as pd
x = pd.Series(
    [2, 4, 4, 5, 7, 9, 12, 41])
print(x.dtype)   # int64
```

Three Questions to Ask

Q1. What is the “typical or common value” of a variable?

Q2. What is the “variability” in the values of a variable?

Q3. Are there unusual or unexpected values in the data?

These three questions organize the whole class, and each one has a small family of statistics attached to it.

Statistics of Location

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Q1. What is the “typical or common value”?

Q1. The “Typical or Common Value”

The **mean**, or sample mean, is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- The **median** has at least half the observations at or below it, and at least half at or above it.
  - Sort the values and pick the middle value.
  - If there are an even number of values, calculate the mean of the two in the middle.
- The **mode** is the most frequently occurring value.
  - The mode works with categorical variables.

WORKED EXAMPLE

$x = 2,\ 4,\ 4,\ 5,\ 7,\ 9,\ 12,\ 41 \quad (n = 8)$

The mean is $\bar{x} = 84/8 = 10.5$.

The two middle values are 5 and 7, so the median is $(5 + 7)/2 = 6$.

The value 4 occurs twice, so the mode is 4.

One observation of 41 pulls the mean far above the median.

Mean, Median, and Mode

R

```r
x <- c(2, 4, 4, 5, 7, 9, 12, 41)
mean(x)     # 10.5
median(x)   # 6

# ignore missing values
mean(c(2, NA, 4), na.rm = TRUE) # 3

# mode() reports storage, not frequency
tab <- table(x)
names(tab)[tab == max(tab)] # "4"
```

Python

```python
x = pd.Series(
    [2, 4, 4, 5, 7, 9, 12, 41])
print(x.mean())     # 10.5
print(x.median())   # 6.0

# pandas skips NaN by default
y = pd.Series([2, None, 4])
print(y.mean())     # 3.0

x.mode().tolist()   # [4]
```

R records a missing value as NA, and a numeric pandas column records it as NaN. R needs `na.rm = TRUE` to omit it; pandas omits it by default. Always report the observed count. Both mode expressions return all tied modes.

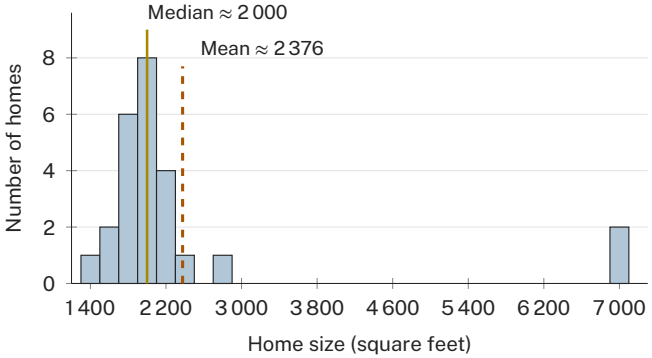
Getting an Accurate “Picture”

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Ten people have an average income of $50,000. Replace the highest earner with someone earning far more. The new mean can be arbitrarily large; the middle two incomes stay the same.

What happens to the median in this example?

A Long Right Tail Moves the Mean

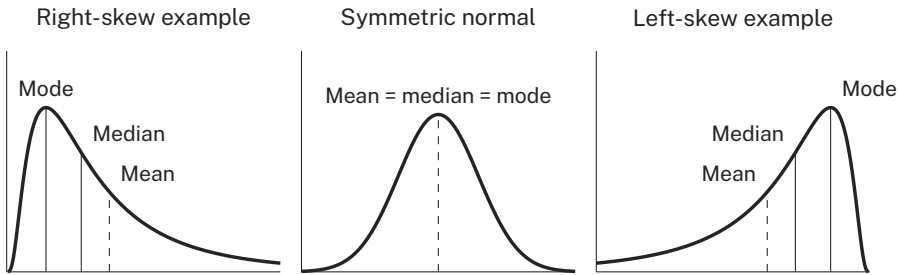


Twenty-five homes. Twenty-three of them lie between 1 400 and 2 800 square feet; two are near 7 000.

Using bin midpoints, the median is about 2 000 and the mean about 2 376. The distant homes pull the mean above the median.

Binned reconstruction after Statology [1].

Skewness Reorders the Statistics



These curves show the common ordering mode < median < mean under right skew, with the reverse under left skew. This ordering is not universal. The normal example is symmetric.

Statistics of Variability or Spread

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Q2. What is the “variability” in the values of a variable?

Q2. What Is the “Variability”? I

The sample variance is

$$s_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

and the standard deviation is

$$s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

What are the units of variance? What about standard deviation?

WORKED EXAMPLE

$x = 2,\ 4,\ 4,\ 5,\ 7,\ 9,\ 12,\ 41$, with $\bar{x} = 10.5$.

$\sum_i (x_i - \bar{x})^2 = 1134$

$s_x^2 = 1134/7 = 162$

$s_x = \sqrt{162} = 12.73$

The divisor $n - 1$ makes this the *sample* variance, the convention Lab 01 uses.

Variance and Standard Deviation

R

```r
x <- c(2, 4, 4, 5, 7, 9, 12, 41)
var(x)   # 162      divisor n - 1
sd(x)    # 12.72792

# Descriptive variance with divisor n
var(x) * (length(x) - 1) / length(x)
# 141.75
```

Python

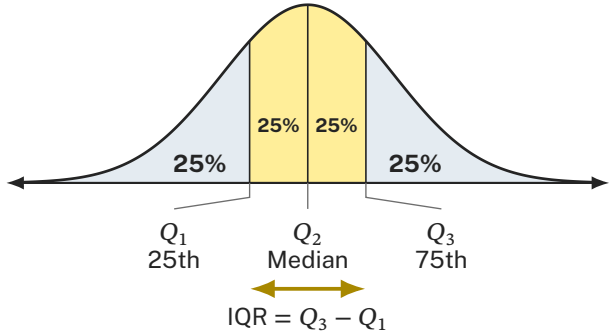
```python
x = pd.Series(
    [2, 4, 4, 5, 7, 9, 12, 41])
print(x.var())        # 162.0  ddof = 1
print(x.std())        # 12.727922...

# Descriptive variance with divisor n
print(x.var(ddof=0))  # 141.75
x.std(ddof=1)         # 12.727922...
```

R's `var`/`sd` and pandas default to the sample divisor $n - 1$. In pandas, `ddof=1` selects this divisor; `ddof=0` selects n. Use the sample version for Lab 01.

Quartiles

- The pth percentile locates the lower $p\%$ of a distribution. Ties and sample interpolation affect the exact fraction.
- Quantiles mark chosen cumulative proportions.
- If data are separated into 4 parts, the quantiles are called *quartiles*:
  - the 1st quartile estimates the 25th percentile;
  - the 3rd quartile estimates the 75th percentile.



Normal example; both tails extend without bound.

Calculate a Type-7 Quartile

Sort the eight values $2, 4, 4, 5, 7, 9, 12, 41$. For a proportion p, calculate the position $h = 1 + (n - 1)p$. Count positions from 1.

- For Q_1, $p = 0.25$ and $h = 2.75$. Positions 2 and 3 both contain 4, so $Q_1 = 4$.
- For Q_3, $p = 0.75$ and $h = 6.25$. Move one quarter of the way from position 6 to position 7.
- Those values are 9 and 12, so $Q_3 = 9 + 0.25(12 - 9) = 9.75$.

This calculation is linear interpolation. The following R and Python calls use this same definition.

Quantiles, and Which Definition

R

```r
x <- c(2, 4, 4, 5, 7, 9, 12, 41)
quantile(x, c(0.25, 0.75), type = 7)
#  25%  75%
# 4.00 9.75          default type 7

fivenum(x)       # hinges, not type 7
# 2.0 4.0 6.0 10.5 41.0
```

Python

```python
x = pd.Series(
    [2, 4, 4, 5, 7, 9, 12, 41])
x.quantile([0.25, 0.75],
           interpolation="linear")
# 0.25  4.00
# 0.75  9.75   linear == R type 7
```

Use R type 7 or pandas linear interpolation for Lab 01. R's `fivenum` and default `boxplot` use hinges, a different rule for dividing a sorted sample. Here their upper hinge is 10.5 rather than 9.75.

Q2. What Is the “Variability”? II

INTERQUARTILE RANGE (IQR)

Difference between the 1st and 3rd quartile.

$$\text{IQR} = Q_3 - Q_1$$

For $x = 2, 4, 4, 5, 7, 9, 12, 41$, $Q_1 = 4$ and $Q_3 = 9.75$, so IQR $= 5.75$.

RANGE

Difference between the maximum and minimum values.

$$\text{range} = x_{(n)} - x_{(1)}$$

For the same data the range is $41 - 2 = 39$, nearly seven times the IQR.

What do you think is an advantage of using the IQR as a measure of spread? Can you think of a disadvantage of using the range?

Outliers and Unknown Values

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Q3. Are there unusual or unexpected values in the data?

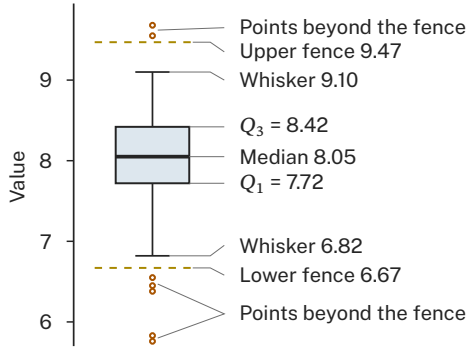
Q3. Unusual or Unexpected Values

Informally, an outlier is an observation unusual enough to prompt investigation of its origin or measurement process.

The boxplot screening rule flags observations outside this interval.

$$\left[\, Q_1 - 1.5 \times \text{IQR},\;\; Q_3 + 1.5 \times \text{IQR} \,\right]$$

A flagged observation is not automatically an error.



Dashed lines are fences; whiskers stop at the last observation inside them.

Applying the Boxplot Rule

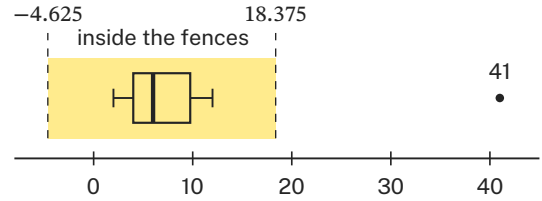
$x = 2,\ 4,\ 4,\ 5,\ 7,\ 9,\ 12,\ 41$

$Q_1 = 4, \quad Q_3 = 9.75, \quad \text{IQR} = 5.75$

$1.5 \times \text{IQR} = 8.625$

Lower fence $4 - 8.625 = -4.625$
Upper fence $9.75 + 8.625 = 18.375$

Only 41 lies outside $[-4.625,\ 18.375]$, so the rule flags exactly one value.



This diagram uses type-7 quartiles. Whiskers end at 2 and 12. R's default boxplot uses hinges, so its box and fences differ; both flag 41.

Missing and Obviously Wrong Values

Look for missing values and obviously wrong ones, such as values of the wrong type.

R

```r
x <- c(2, 4, 4, 5, 7, 9, 12, 41, NA)
boxplot.stats(x)$out
# 41
which(is.na(x))
# 9
sum(is.na(x))
# 1
```

Python

```python
x = pd.Series(
    [2, 4, 4, 5, 7, 9, 12, 41, None])
q1, q3 = x.quantile([0.25, 0.75],
                    interpolation="linear")
iqr = q3 - q1
print(x[(x < q1 - 1.5 * iqr) |
        (x > q3 + 1.5 * iqr)])
# 7    41.0
x.index[x.isna()].tolist()
# [8]
```

`boxplot.stats` builds its fences from hinges, so they can differ slightly from fences built on `quantile` type 7. Here both flag only 41.

Global Summaries in R

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R

```r
x <- c(2, 4, 4, 5, 7, 9, 12, 41)
summary(x)
#  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#  2.00    4.00    6.00   10.50    9.75   41.00
table(c("a", "b", "a", "c"))
# a b c
# 2 1 1
```

`summary()` reports the minimum, the two quartiles, the median, the mean, and the maximum in one call.

`table()` counts the observations in each category. A frequency table is the summary for a categorical variable.

Global Summaries in Python

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Python

```
x = pd.Series(
    [2, 4, 4, 5, 7, 9, 12, 41])
print(x.describe().round(2))

s = pd.Series(list("abac"))
print(s.value_counts())

print(x.agg(["mean", "median", "std"]))
```

Printed

```
count     8.00
mean     10.50
std      12.73
min       2.00
25%       4.00
50%       6.00
75%       9.75
max      41.00
a    2
b    1
c    1

mean      10.500000
median     6.000000
std       12.727922
```

R's `summary()` reports the mean; pandas' `describe()` adds the standard deviation, computed with `ddof=1`, which matches R's `sd()`.

What to Take Away

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Location

Mean, median, mode. Only the mode applies to nominal categories.

Variability

Variance, standard deviation, interquartile range, and range describe different aspects of spread.

Outliers

Flag possible outliers with the boxplot rule, then investigate their source and context.

Interpretation

Choose summaries for the question. Check plots, missingness, units, and software defaults.

Summary-Statistics Warm-Up

Practice in R or Python and generate basic data summaries.

Use a local or cloud environment. Compare your output with the worked example, then begin integrated Lab 01.

One Row Is One Sensor Reading

```
record_id,site,temperature_c,oxygen_percent,depth_m
r01,A,2,70,1
r02,A,4,80,2
r03,A,4,80,2
r04,A,5,85,3
r05,B,7,90,3
r06,B,9,95,4
r07,B,12,105,4
r08,B,41,130,10
r09,A,,75,2
```

Save as `practice.csv`.

Temperature is in degrees Celsius, oxygen saturation in percent, and depth in metres. Site labels define groups; record IDs identify rows.

The blank field is missing, not zero. Lab 01 uses at least 100 observations from an approved source.

Read the CSV and Select Measurements

R

```r
observations <- read.csv("practice.csv", stringsAsFactors = FALSE)
numeric_cols <- c("temperature_c", "oxygen_percent", "depth_m")
dir.create("results", showWarnings = FALSE)
```

Python

```python
from pathlib import Path
import matplotlib.pyplot as plt
import pandas as pd
observations = pd.read_csv("practice.csv")
numeric_cols = ["temperature_c", "oxygen_percent", "depth_m"]
Path("results").mkdir(exist_ok=True)
```

Keep the script and CSV in one folder. The demonstration follows the complete example in the notes, which writes one summary row for each of these columns.

Check the Output Before Adapting It

Overall temperature

- 8 observed, 1 missing
- Mean 10.5, median 6
- Sample SD 12.73
- $Q_1 = 4$, $Q_3 = 9.75$, IQR 5.75
- Type-7 fences flag 41 (row `r08`)

Compare the sites

- A has 4 observed temperatures, mean 3.75
- B has 4 observed temperatures, mean 17.25
- The value 41 contributes to the difference
- A statistical flag alone cannot establish a measurement error

Save Evidence That Can Be Checked

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1. Save the summary, group comparison, and flagged rows as CSV tables.
2. Make a temperature histogram and a temperature–oxygen scatter plot. Label variables, units, and the number of observations used.
3. Save the plots as PDFs. Record the software environment and the command that runs the script.
4. Restart and run the full analysis. Check that it recreates every result cited in the report.

For Lab 01, also record the dataset publisher, URL, licence, access date, selected station and period, and missing-value codes. Keep the original download unchanged.

Commit, Check, and Submit Privately

1. Create a private GitHub repository and give the instructor access. Store the lab under `labs/lab-01/`.
2. A commit records a snapshot. Use the browser's *Commit changes*, or commit and push from local Git. Record meaningful stages as you work.
3. Open the final commit on GitHub and check its files. Record its full SHA and commit URL in `labs/lab-01/README.md`, then commit and push before 16:05 Halifax time on September 17.

The notes give the steps and commands. Corrections are new commits. Publish the same submitted commit and history only after the deadline and the instructor's release.

References

[1] Z. Bobbitt. *How Do Outliers Affect the Mean?* Statology. Jan. 29, 2020. URL: https://www.statology.org/how-do-outliers-affect-the-mean/.